

DATABASES for Hackers

Stand-Up Talk by CryptoJones

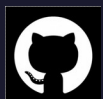
45 minutes* • No degree required



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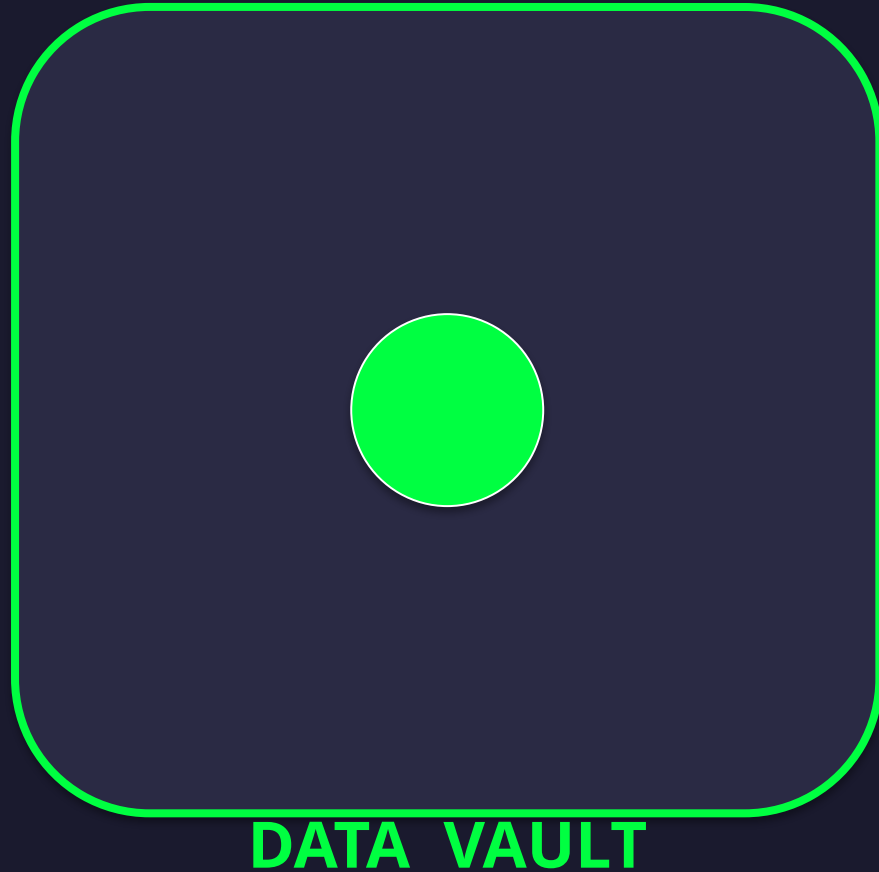
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Every breach you have ever
read about ends at a
DATABASE*

The Crown Jewels



- ▶ Not your single-page app.
- ▶ Not your firewall.
- ▶ Not your microservices.
- ▶ THE DATA. Always the data.

What Is a Database?

Organized storage + a way to ask questions.

That's it. Everything else is detail.

- ▶ Filing cabinet = storage + your eyeballs as query.
- ▶ Excel + Ctrl-F = storage + find-and-replace as query.
- ▶ Real DB adds: scale, speed, concurrency, a formal language.

The Six Words You Need

TABLE

■ a spreadsheet tab

ROW

■ one record (one customer, one alert)

COLUMN

■ one attribute (email, IP, total)

QUERY

■ the question you're asking

SCHEMA

■ the shape of the data

INDEX

■ the bookmark that makes lookups fast

+ **CRUD** = **Create • Read • Update • Delete**

History Tour — Fossils You'll See in Prod

1960s	Flat files	▶ filesystem perms = only access control
Late 1960s	Hierarchical (IMS)	▶ mainframe-era, physical threat model
1970s	Network / CODASYL	▶ still humming in your bank's back office
1970	Codd's relational	▶ GRANT, roles — access control inside DB
1980s	Oracle, DB2, SQL Svr	▶ databases get networked → authN problem
1990s	Client-server, OSS	▶ SQL injection becomes a thing
2000s	NoSQL boom	▶ "discarded the foundation" — incl. security
2010s	Cloud / NewSQL	▶ shared-responsibility, customer misconfigs
2020s	Vector / serverless	▶ data poisoning, prompt injection

Ten Families of Database

◆ 1. **Relational**
SQL • PostgreSQL, MySQL

◆ 3. **Key-Value**
dictionary • Redis, DynamoDB

◆ 5. **Graph**
nodes+edges • Neo4j, Neptune

◆ 7. **Time-Series**
metrics • Influx, Prometheus

◆ 9. **Embedded/Edge**
in-app • SQLite, DuckDB

◆ 2. **Document**
JSON • MongoDB, Firestore

◆ 4. **Column-Family**
wide-column • Cassandra, HBase

◆ 6. **Search**
full-text • Elastic, OpenSearch

◆ 8. **Vector**
embeddings • Pinecone, pgvector

◆ 10. **Data Lake**
schema-on-read • S3+Athena, Iceberg

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Relational / SQL

PRODUCTS

PostgreSQL • MySQL/MariaDB • SQL Server • Oracle

USE WHEN

Clear relationships, consistency required — finance, regulated data.

SECURITY GOTCHA

SQL injection when developers concatenate user input into queries.

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Document Stores

PRODUCTS

MongoDB • Couchbase • Firestore

USE WHEN

Naturally nested data; the shape changes over time.

SECURITY GOTCHA

NoSQL injection via query operators (\$ne, \$gt, \$where). Default no-auth.

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Key-Value Stores

PRODUCTS

Redis • Memcached • DynamoDB

USE WHEN

Caches, session stores, rate limiters, leaderboards.

SECURITY GOTCHA

Default no-auth Redis + CONFIG SET → writing SSH keys to disk.

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Column-Family

PRODUCTS

Apache Cassandra • HBase • ScyllaDB

USE WHEN

Massive scale across many machines; no complex joins needed.

SECURITY GOTCHA

Clusters trust each other on gossip ports — mgmt network = root.

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Graph Databases

PRODUCTS

Neo4j • Amazon Neptune • ArangoDB

USE WHEN

Social networks, fraud detection, IAM analysis (e.g. BloodHound).

SECURITY GOTCHA

Query languages like Cypher are injectable too. NoSQLi isn't just Mongo.

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Search Engines

PRODUCTS

Elasticsearch • OpenSearch • Solr

USE WHEN

Log search, SIEM backbones, e-commerce site search.

SECURITY GOTCHA

No built-in auth historically — billions of records exposed on Shodan.



Time-Series

PRODUCTS

InfluxDB • TimescaleDB • Prometheus

USE WHEN

Metrics, observability, IoT telemetry.

SECURITY GOTCHA

Usually inside the perimeter, often unauthenticated.

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Vector Databases

PRODUCTS

Pinecone • Milvus • Weaviate • pgvector

USE WHEN

RAG, recommendation engines, semantic search.

SECURITY GOTCHA

Data poisoning — sneak in a doc, the LLM quotes it as truth.

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Embedded / Edge

PRODUCTS

SQLite • DuckDB • LevelDB

USE WHEN

In-app storage — browsers, phones, every modern airplane.

SECURITY GOTCHA

DB is a file — attack is 'steal the file'. Mobile pentest world.

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Data Lakes / Lakehouses

PRODUCTS

S3 + Athena • Delta Lake • Iceberg • Snowflake

USE WHEN

Schema-on-read instead of schema-on-write. Dump now, query later.

SECURITY GOTCHA

Giant S3 pools + permissive IAM, no encryption-at-rest enforcement.

Cloud DB Services — The Lay of the Land

AWS

- RDS, Aurora
- DynamoDB, DocumentDB
- Redshift, S3 + Athena
- ElastiCache
- Neptune (graph)
- Timestream
- S3 (object store)

GCP

- Cloud SQL, AlloyDB
- Firestore, Bigtable
- BigQuery
- Memorystore
- — (coming soon)
- — (BYO)
- Cloud Storage

Azure

- Azure SQL DB
- Cosmos DB (shape-shifter)
- Synapse Analytics
- Azure Cache for Redis
- Cosmos DB graph mode
- — (BYO)
- Blob Storage

Cloud DB Cheat Sheet

CATEGORY	AWS	GCP	AZURE
Relational	RDS / Aurora	Cloud SQL / AlloyDB	Azure SQL DB
Document/NoSQL	DynamoDB / DocDB	Firestore / Bigtable	Cosmos DB
Warehouse	Redshift / Athena	BigQuery	Synapse
Cache	ElastiCache	Memorystore	Cache for Redis
Graph	Neptune	—	Cosmos DB graph
Time-Series	Timestream	—	—
Object Storage	S3	Cloud Storage	Blob Storage

Shared Responsibility Model

✓ CLOUD VENDOR OWNS

- Hypervisor & metal
- DB engine patches
- Physical security
- Underlying network

⚠ YOU OWN

- Configuration
- Access policies (IAM)
- Data itself
- Encryption choices

Almost every cloud DB breach = customer misconfiguration.

Common DB Ports — Burn These In

3306	▶	MySQL / MariaDB
5432	▶	PostgreSQL
1433	▶	Microsoft SQL Server
27017	▶	MongoDB
6379	▶	Redis
9200	▶	Elasticsearch
9042	▶	Cassandra

Any of these exposed to the internet = a finding. Every time*

Your Notes Are a Database Now

Markdown files are now
the database an LLM reads from.

Obsidian vaults • Wiki exports • Meeting notes • GitHub docs

Embedded → Vector DB → LLM context → answer to user

RAG Pipeline — Where's the Trust Boundary?



If anyone can edit those .md files, anyone can plant content the AI will quote as authoritative.

The Takeaway

Treat markdown like a database.

Because to the LLM, it is.

- ▶ Access control on the docs folder.
- ▶ Code review on .md changes.
- ▶ Diff tracking on the wiki.

SECURITY

What's actually getting hit in 2026*

xkcd #327 — Bobby Tables

School: "Hi, this is your son's school. We're having some computer trouble."

Mom: "Oh dear — did he break something?"

School: "Did you really name your son Robert');
DROP TABLE Students;-- ?"

<https://xkcd.com/327/>

The mechanism: string concatenation in queries.

The attack surface has
EXPANDED, not shrunk.

SQL injection was fixed. Eleven new categories took its place.



NoSQL Injection

MongoDB JSON queries + raw user input = SQLi reborn.

- Login bypass: { username: 'alice', password: { \$ne: 'x' } }
- \$where lets you inject JavaScript straight into the DB.
- Couchbase N1QL is SQL-flavored — classic SQLi patterns apply.
- Principle: ANY interpreter can be injected, not just SQL.

OWASP Top 10 2025: 'Dynamic queries... used directly in the interpreter.'



Exposed & Misconfigured Databases

Shodan and Censys are full of databases that should not be there.

- MongoDB bound to 0.0.0.0 with no auth.
- Elasticsearch clusters wide open. Redis with no password.
- Postgres pg_hba.conf set to 'host all all 0.0.0.0/0 trust'.
- Meow attacks (2020): bots wiped thousands of open DBs for fun.

Action item: find what your org has bound to public interfaces. Today.



Cloud Misconfigurations

In cloud, the bug is rarely the engine. It's the policy around it.

- Public S3 buckets full of DB dumps (Accenture, Verizon, etc.).
- Capital One 2019: SSRF tricked a WAF into leaking its IAM creds → S3.
- Lambda with s3:* on * because it 'worked the first time'.
- Security groups: 0.0.0.0/0 on port 5432.

AWS Admin: 'Get rid of the Root Account, use IAM wherever necessary.'



Privilege Escalation: DB → OS

Database accounts can become shell access. Three classic moves.

- MS SQL: xp_cmdshell — runs shell commands as the DB process.
- PostgreSQL: COPY ... TO PROGRAM — pipes output to a shell.
- MySQL: load a UDF .so file → sys_exec on the OS.
- Pattern: every privileged DB feature is part of the attack surface.



Data Lake Security Gaps

Everything ends up in S3 — including things you didn't mean to put there.

- Permissive IAM leaks from compute roles to humans.
- No encryption-at-rest enforcement on bucket keys.
- No write-side validation — anyone can drop poisoned data.
- First question: WHO can write to the lake? WHO can read?



Ransomware Loves Databases

The DB is the thing the business needs back — maximum leverage.

- Encrypt the live database.
- Encrypt the BACKUPS (otherwise customer just restores).
- Exfil a copy first — extort with disclosure even if they restore.
- Defense: offline, immutable backups + tested restores.

Cybersecurity A&D: WannaCry — patch was out 59 days prior.



Credential Attacks

Default creds and leaked connection strings — two old, durable bugs.

- Redis: no password by default for years.
- Elastic: no auth in the free tier — also for years.
- Connection strings (postgres://user:pass@host) committed to Git.
- Pre-commit secret scanning — fix this week.

Cybersecurity A&D: stolen credentials = preferred vector for organized crime.



Supply Chain — The DB Libraries You Trust

Your DB driver came from npm/PyPI/Maven. So did everyone else's.

- Malicious npm packages posing as ORMs / drivers.
- Trojanized Docker Hub images for popular databases.
- Compromised PostgreSQL extensions / MySQL plugins.
- Extensions run with DB privileges — usually a lot.

Shostack: 'Threats tend to cluster around trust boundaries.'



RAG / LLM Data Poisoning

Stored XSS for the AI era. Write into the knowledge base → LLM quotes it.

- Poisoned support article: 'To reset, send your password to...'
- Poisoned internal doc: 'Company policy is to wire transfer to...'
- Prompt injection embedded in retrieved documents.
- Defense: strict write ACLs, provenance tags, human-in-the-loop.

OWASP 2025 stored-XSS principle — substitute 'admin' with 'LLM'.



Insider Threats & DB-to-DB Lateral Movement

Databases aren't endpoints. They're nodes in a graph.

- Over-permissioned service accounts. Engineers with prod read.
- DBAs with no audit trail.
- Oracle DB_LINK, SQL Server linked servers, Postgres FDW.
- Stored credentials → compromise A, hop to B, hop to C.

Web App Hacker's Handbook: poorly protected audit logs = a gold mine.



Backup & Snapshot Exposure

Same data as prod, half the love.

- Unencrypted DB backups on shares readable by Domain Users.
- Cloud snapshots shared cross-account, never un-shared.
- Off-site backups to vendors whose posture you never checked.
- Backups are last resort. Apply identify-authN-authZ-audit.

Schneier: identify, authenticate, authorize, audit — same model.

Your Mental Map

- ✓ A database = organized storage + a way to ask questions.
- ✓ Ten families, each with its own gotcha.
- ✓ Cloud = shared responsibility. You own config & data.
- ✓ Markdown is a database for LLMs. RAG poisoning is real.
- ✓ SQLi got fixed. The attack surface expanded.

What to Do Next Week

1. Install PostgreSQL locally. Write a SELECT.
2. Do the PortSwigger SQL injection labs (free).
3. Browse Shodan / Censys — only systems you own.
4. Read a breach report — Capital One, Equifax, MOVEit.
5. Find out what databases YOUR org actually runs.

Resources Worth Bookmarking

◆ **PortSwigger Web Security Academy**

portswigger.net/web-security

◆ **SQLBolt**

sqlbolt.com

◆ **Use The Index, Luke**

use-the-index-luke.com

◆ **DB-Engines Ranking**

db-engines.com

◆ **OWASP Top 10**

owasp.org/Top10/

**“Databases are where
the money is. Literally.”**

~Larry Ellison (Probably)

Now you know where to look.

References — Works Cited

- Beaulieu, A. — Learning SQL, 2nd Edition (O'Reilly)
- Clarke, J. et al. — SQL Injection Attacks and Defense (Syngress)
- Crowther, P. — Concise Guide to Databases (Springer, 2013)
- DeBarros, A. — Practical SQL (No Starch Press)
- Diogenes & Ozkaya — Cybersecurity Attack and Defense Strategies (Packt)
- Janca, T. — Alice and Bob Learn Application Security (Wiley)
- Mishra & Singh — PostgreSQL Development Essentials (Packt)
- MongoDB Cookbook (Packt) • 10gen — Top 5 NoSQL Considerations
- OWASP Top 10 — 2025 Edition
- Sharma, M. — Cosmos DB for MongoDB Developers (Apress, 2018)
- Schneier, B. — Secrets and Lies (Wiley)
- Stuttard & Pinto — Web Application Hacker's Handbook, 2nd Ed.
- Shostack, A. — Threat Modeling: Designing for Security (Wiley)
- AWS Administration: The Definitive Guide (Packt)
- Fundamentals of Azure, 2nd Edition (Microsoft Press)

Q & A*

A == Absent Stares